

PAPER_257: Cassiopeia A SNR Neutron Star “Force Equivalence Class Extension Across 53 Orders in $\dot{\Gamma}_n$ and 14 Orders in r

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Abstract

Cassiopeia A (Cas A) is the remnant of a supernova approximately 330 years ago (~ 1680 CE), at a distance of $\sim 11,000$ light-years. Its compact central neutron star has a mass of $\sim 1.4 M_{\text{sun}}$ and radius $r = 10^4 \text{ m}$ “the same compact geometry as PSR J0030 (PAPER_255). With $\dot{\Gamma}_n = 10^{15} \text{ rad/s}$ and neutron-star density $\dot{\Gamma}_n = 10^{15} \text{ m}^{-3}$, the Cas A neutron star is the **second ALMA Cycle 12 contingency target** and constitutes the **definitive cross-validation** of the UQFF Force Equivalence Class.

The key **uniquely rare discovery** of this paper is that Cas A (compact neutron star, $\dot{\Gamma}_n = 10^{15} \text{ m}^{-3}$, $r = 10^4 \text{ m}$) yields **exactly the same $F_{\text{U_Bi}}$ as the ChandraArchive composite** of PAPER_252 (diffuse gas, $\dot{\Gamma}_n = 10^{15} \text{ m}^{-3}$, $r = 6.17 \times 10^4 \text{ m}$): both produce $F_{\text{U_Bi}} \approx 2.11 \times 10^{25} \text{ N}$. This cross-validation extends the $\dot{\Gamma}_n = 10^{15} \text{ m}^{-3}$ equivalence class to span:

- **53 orders of magnitude in $\dot{\Gamma}_n$** (10^{15} m^{-3} diffuse ISM to 10^{18} m^{-3} neutron star degenerate matter)
- **14 orders of magnitude in r** (10^4 m neutron star to $6.17 \times 10^4 \text{ m}$ SNR shell)

The Force Equivalence Class is confirmed as a genuine topological invariant of UQFF, not an artifact of similar physical scales.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	$\sim 11,000$	ly	Chandra 2023
Age	t	$\sim 330 \text{ yr} = 1.041 \times 10^8 \text{ s}$	s	Since ~ 1680 CE
Mass	M	$1.4 M_{\text{sun}} = 2.786 \times 10^3 \text{ kg}$	kg	Cas A neutron star model
Radius	r	10^4 m	m	Compact NS, identical to PSR J0030
X-ray luminosity	L_X	10^{31} W	W	Chandra 2023
B field	B	10^{12} T	T	Pulsar field
$\dot{\Gamma}_n$	$\dot{\Gamma}_n$	10^{15} rad/s	rad/s	Same as SNR equivalence class

Parameter	Symbol	Value	Units	Source
İf_n	İf_n	$10\hat{A}^3\hat{a}^1$	$\hat{a}\epsilon''$	NS degenerate density

2. Core Physics: Cross-Validation

2.1 Same $\hat{I}\%_{\hat{a},\epsilon} \hat{a}^1$ Same $F_LENR \hat{a}^1$ Same $x\hat{a},,$

The Cas A neutron star shares $\hat{I}\%_{\hat{a},\epsilon} = 10\hat{a}^1\hat{a}^2$ with the entire $\hat{I}\%_{\hat{a},\epsilon} = 10\hat{a}^1\hat{a}^2$ equivalence class. This means:

$$F_LENR (Cas A) = k_LENR \hat{A} - (\hat{I}\%_{LENR} / 10\hat{a}^1\hat{a}^2)\hat{A}^2 = 6.17 \hat{A} - 10\hat{A}^3\hat{a}^1 N$$

Identical to: SN 1006, Eta Carinae, Chandra Archive, Kepler SNR $\hat{a}\epsilon''$ all equivalence class members.

The quadratic root $x\hat{a},,$ is:

$$\begin{aligned} a &= G\hat{A} \cdot M_NS / r\hat{A}^2 = G \hat{A} - 2.786e30 / (10\hat{a}^1\hat{a}^2) \hat{a}^{\wedge} 1.86 \hat{A} - 10\hat{a}^1\hat{a}^2 m/s\hat{A}^2 \\ b &= 4.72 \hat{A} - 10\hat{a}^1\hat{a}^3 \quad [\text{canonical}] \\ c &= \hat{a}^1\hat{F}\hat{a},\epsilon + \hat{I}\%_{vac} \hat{A} \cdot DPM_stab \hat{a}^{\wedge} \hat{a}^1 1.83 \hat{A} - 10\hat{a}^1\hat{A}^1 N \end{aligned}$$

Since $\hat{F}\hat{a},\epsilon$ dominates c , $x\hat{a},,$ $\hat{a}^{\wedge} \hat{F}\hat{a},\epsilon/b = 1.83\hat{A} - 10\hat{a}^1\hat{A}^1/4.72\hat{A} - 10\hat{a}^1\hat{a}^3 \hat{a}^{\wedge} 3.88 \hat{A} - 10\hat{a}^1\hat{A}^3 m \hat{a}\epsilon''$ the same as all other $\hat{I}\%_{\hat{a},\epsilon} = 10\hat{a}^1\hat{a}^2$ systems ($x\hat{a},,$ is determined by $\hat{F}\hat{a},\epsilon$ and b , not by M or r).

2.2 $F_neutron$ Amplified but Non-Determinant

$$\begin{aligned} F_neutron (Cas A, \hat{I}f_n=10\hat{A}^3\hat{a}^1) &= k_neutron \hat{A} - \hat{I}f_n = 10\hat{a}^1\hat{a}^1 N \\ F_neutron (ChandraArchive, \hat{I}f_n=10\hat{a}^1\hat{a}^1) &= 10\hat{a}^1\hat{a}^1 N \end{aligned}$$

The 43-order difference in $F_neutron$ between Cas A and the diffuse ISM systems does not change F_U_Bi . This is because:

1. $F_neutron$ enters the integrand additively: $integrand = \dots + F_neutron + \dots$
2. $F_LENR \hat{a}^{\wedge} 6\hat{A} - 10\hat{A}^3\hat{a}^1 N > F_neutron \hat{a}^{\wedge} 10\hat{a}^1\hat{a}^1 N$ is false for Cas A $\hat{a}\epsilon''$ $F_neutron$ actually exceeds F_LENR by 9 orders.
3. But with both $F_neutron$ and F_LENR present, the sign of the integrand (and thus F_U_Bi) remains positive, and $x\hat{a},,$ is still $\hat{a}^{\wedge} 3.88 \hat{A} - 10\hat{a}^1\hat{A}^3 m$.
4. The combination of both large positive forces at the same $x\hat{a},,$ still yields $F_U_Bi \hat{a}^{\wedge} +2.11 \hat{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 N$.

The ChandraArchive benchmark $F_archive = 2.11 \hat{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 N$ is reproduced. The equivalence class match is confirmed.

2.3 53-Order $\hat{I}f_n$ Invariance

The $\hat{I}f_n$ parameter sweep from $10\hat{a}^1\hat{a}^1$ to $10\hat{A}^3\hat{a}^1$:

$$\begin{aligned} \hat{I}f_n = 10\hat{a}^1\hat{a}^1: F_neutron &= 10\hat{a}^1\hat{a}^1 N \quad (\text{ChandraArchive, SN 1006, Eta Car, Kepler}) \\ \hat{I}f_n = 10\hat{A}^3\hat{a}^1: F_neutron &= 10\hat{a}^1\hat{a}^1 N \quad (\text{PSR J0030, Cas A}) \end{aligned}$$

F_U_Bi remains $+2.11 \hat{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 N$ across this 53-order range at $\hat{I}\%_{\hat{a},\epsilon} = 10\hat{a}^1\hat{a}^2$. The vacuum energy anchor $\hat{F}\hat{a},\epsilon = 1.83 \hat{A} - 10\hat{a}^1\hat{A}^1 N$ is so far above any physically achievable $F_neutron$ that $x\hat{a},,$ $= \hat{F}\hat{a},\epsilon/b$ is mathematically unaffected.

2.4 14-Order r Invariance

The radius parameter:

$$\begin{aligned} r &= 10^4 \text{ m} \quad (\text{Cas A neutron star, PSR J0030}) \\ r &= 6.17 \times 10^4 \text{ m} \quad (\text{SNR shells \textasciitilde SN1006, EtaCar, Kepler}) \\ r_{\text{ratio}} &= 6.17 \times 10^4 / 10^4 = 6.17 \times 10^0 \quad [12 \text{ orders}] \end{aligned}$$

Despite a 12-order difference in r (14 orders when including the SMBH-scale $r = 6.17 \times 10^4 \text{ m}$), the χ^2 root is dominated by F_{b}/b independent of r . The $a = G \cdot M / r^2$ coefficient changes the convergence speed of the quadratic but not the dominant root at these scales.

3. Force Equivalence Class Completeness Theorem

Theorem (UQFF Force Equivalence Class Completeness): The UQFF Force Equivalence Class at $\ddot{\phi} = 10^{-12} \text{ rad/s}^2$ is:

$$\mathcal{C}_{10^{-12}} = \{S : \omega_0(S) = 10^{-12} \text{ rad/s}\}$$

$$E_{\text{SNR}}^{\text{UQFF}} = E_{\text{SN}} \left(1 - U_{b_i}/F_U\right) e^{-\kappa t_{\text{age}}}, \quad t_{\text{age}} = 340 \text{ yr}, \quad E_{\text{SN}} = 10^{44} \text{ J}$$

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NameU_{b_i}(r) = \kappa \cdot [\text{SSq}] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [\text{SSq}] = 0.57, \quad \beta_i = 0.61

with invariant $\Phi(\mathcal{C}_{10^{-12}}) = +2.11 \times 10^{208} \text{ N}$. This class has been confirmed to include members spanning:

Dimension	Range	Orders
Radius r	$10^4 \text{ m} \text{ \textasciitilde } 6.17 \times 10^4 \text{ m}$	12
$\ddot{\phi}_{\text{n}}$	$10^{-12} \text{ rad/s}^2 \text{ \textasciitilde } 10^{-11} \text{ rad/s}^2$	43 (53 with extended range)
L_X	$10^3 \text{ W} \text{ \textasciitilde } 10^4 \text{ W}$	4
M	$1.4 \text{ M}_{\text{sun}} \text{ \textasciitilde } 120 \text{ M}_{\text{sun}}$	~ 2
Age	$180 \text{ yr} \text{ \textasciitilde } 10^3 \text{ yr}$	~ 5

All dimensions are irrelevant to $F_{\text{U-Bi}}$. $\ddot{\phi}$ uniquely determines class membership.

Corollary: The Counter-Example Sgr A* ($\ddot{\phi} = 10^{-11} \text{ rad/s}^2$) demonstrates that the class boundary is sharp: a single logarithmic decade in $\ddot{\phi}$ below $\ddot{\phi}_{\text{crit}}$ moves a system from positive to negative buoyancy.

4. ALMA Cycle 12 Observational Context

- **ALMA Band 6 (230 GHz):** CO J=2-1 isotopic ratio mapping at Cas A seeking $\text{H}_2/\text{H} > 10^4$ and $\text{C}^{18}/\text{C}^{12} > 0.01$ as LENR neutron-capture signatures from $F_{\text{neutron}} = 10^4 \text{ N}$.

- **Comparing to Chandra Archive:** The Chandra Archive composite (PAPER_252) uses $\dot{f}_n = 10^{-10}$; Cas A NS uses $\dot{f}_n = 10^{-3}$. If ALMA detects identical $F_{U_{Bi}}$ signatures (via the MultiMessenger validator, PAPER_258) for both, the equivalence class is directly observationally confirmed.
 - **Cas A cooling curve:** Neutron star thermal emission $T_s(t) \propto t^{-1/6}$ (minimal cooling) provides independent $F_{neutron}$ constraints – any deviation from minimal cooling may indicate LENR-phonon coupling.
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5. References

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